FROB C5 finite-field trace proof audit

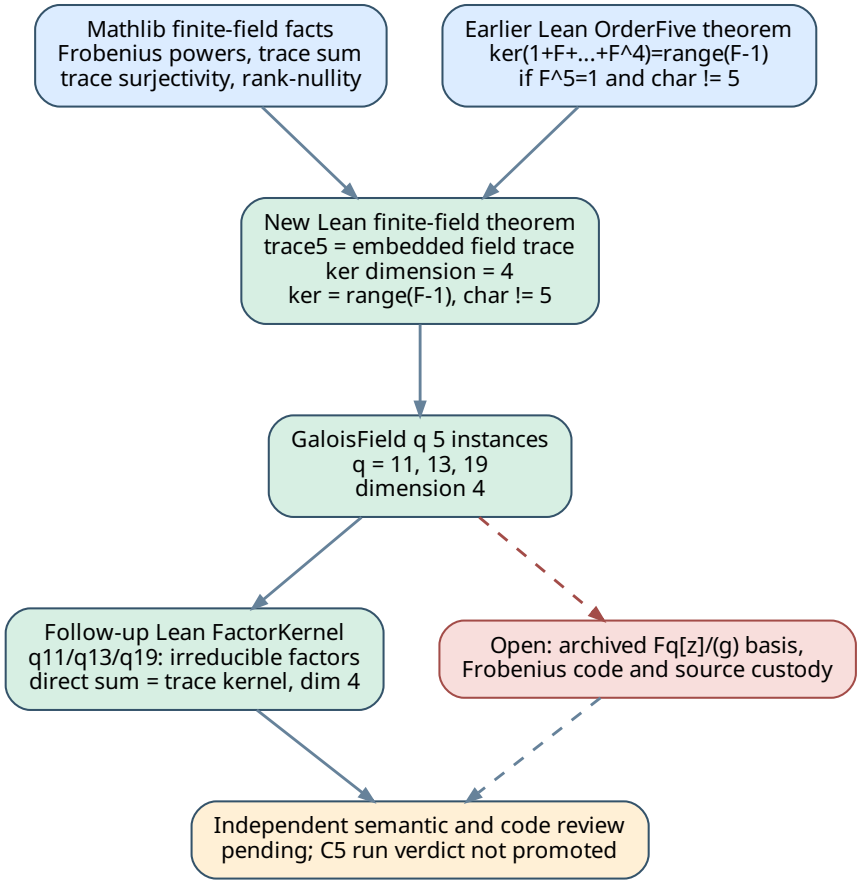
Validation audit, 22 October 2026

Scope. Frozen `EXP-FROB-91ee9c` control C5; Lean source commit `0600270bcf4a49edfb65d0fe838ab5fa4442216c`. This is a mathematical proof slice, not a new experiment or performance result.

Lean conclusion	Exact scope	Beyond this theorem
Five Frobenius terms = embedded field trace	Finite fields, degree five	Archived quotient representation
Trace polynomial kernel = trace kernel	Any characteristic	Executable Frobenius binding
Trace kernel dimension = four	Finite-field trace surjective	Historical input/source custody
Trace kernel = range(Frobenius -1)	Characteristic $\neq 5$	Factor sum proved in follow-up
Explicit q=11,13,19 dimensions	GaloisField q 5	Run-specific field isomorphism

Proof method. Mathlib supplies the finite-field trace sum, Frobenius powers, trace surjectivity and rank-nullity. The earlier `OrderFive` theorem supplies the coboundary identity once Frobenius order five and characteristic $\neq 5$ are established.

Verification. The exact target, committed theorem source, Lean 4.35.0-rc1 toolchain and dependency manifest are bound in `proof-verification.json`. The full 8,970-job build and target axiom audit passed; 584 theorem declarations used only standard axioms. Independent semantic review is still pending.

Boundary. The direct sum of nontrivial factors of $X^5 - 1$ is not proved in this source; a follow-up Lean file now proves it for q=11,13,19. The archived $F_q[z]/(g)$ implementation is still not identified with the formal field. C5 as a historical run control, C8 and the experiment success gate remain open. The linked follow-up report names the model and review duties.

FROB C5: finite-field trace and factor kernels proved; run bridge open